

ZEISS AI IOL Calculator

Software version 1.1

Document set



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Instructions for use

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Table of Contents

1	Notes on the instructions for use	5
1.1	Designations.....	5
1.2	Scope.....	5
1.3	Purpose and safe-keeping of the documentation	5
1.4	Questions and comments	5
1.5	Representations in this document	6
1.5.1	Representations of safety and warning instructions	6
1.5.2	Representations in all text sections	6
1.5.3	Representations in action sequences	7
2	Safety notes	8
2.1	Target user group.....	8
2.2	Target patient group	8
2.3	Area of use	8
2.3.1	Intended use	8
2.3.2	Indications.....	8
2.3.3	Contraindications	8
2.4	Operator responsibilities and obligations	9
2.4.1	Operating staff	9
2.4.2	Messages to manufacturer and authorities	9
2.4.3	Data protection and information security	10
2.4.4	Place of installation and network.....	10
2.5	Patient information	10
2.6	Product support period	10
3	Software description	11
3.1	Explanation of symbols.....	11
3.2	Conformity	11
3.3	Functional description	12
4	Troubleshooting.....	14

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1 Notes on the instructions for use

1.1 Designations

In these instructions for use, ZEISS AI IOL Calculator is also referred to as "software".

1.2 Scope

These instructions for use are valid for ZEISS AI IOL Calculator software version 1.1.

1.3 Purpose and safe-keeping of the documentation

These instructions for use describe the product's safety equipment, functions, and performance parameters. They describe safe use.

Correct use of the product is paramount for safe and successful operation.

Use the following link to download the electronic instructions for use (eIFU) at any time:

www.zeiss.com/ifu

Contact your local sales representative to obtain a hard copy of the instructions for use at any time.

Action

- ▶ Carefully read these instructions for use prior to using the product.
- ▶ Make sure that all users have access to the instructions for use at all times.
If you received the instructions for use in electronic form, save them in a storage location accessible to all users.

1.4 Questions and comments

Action

- ▶ If you have any questions or comments about these instructions for use or the software itself, please contact ZEISS Service.

You can find the ZEISS representative for your country online on the following website: www.zeiss.com/med

1.5 Representations in this document

1.5.1 Representations of safety and warning instructions

WARNING!

This is a warning label that warns of hazards where death or serious injury may occur if not avoided.

The warning label states possible consequences.

- This is a measure that can be used to prevent hazards.

CAUTION!

This is a warning label that warns of hazards where injury may occur if not avoided.

The warning label states possible consequences.

- This is a measure that can be used to prevent hazards.

NOTE

This is a warning label that warns of hazards where property damage may occur if not avoided.

The warning label states possible consequences.

- This is a measure that can be used to prevent hazards.

1.5.2 Representations in all text sections

- This is a list.
 - This is a second-level list.

This is a cross-reference: see Representations in all text sections [► 6].

This is an **emphasis**.

This is a `software code` or a `program text`.

Names of software dialogs, fields or menus as well as software messages are marked with quotation marks:

- "View" menu
- "Do you want to save the settings"

The steps in menu and file paths are separated by slashes:

- "File / Save as"
- "My files / Documents"

Keys and buttons are marked with square brackets:

- [Next] button
- [Alt] key

TIP: Tips are marked with a highlighted "TIP:".

INFO: Important additional information is marked with "INFO:".

1.5.3 Representations in action sequences

Action sequences are structured as follows:

- Prerequisite
- Procedure
- Result

The representation of the action sequences may vary. There are the following representations of action sequences:

Defined action sequence

Complex actions that must be carried out one after the other in a clear sequence (step by step) are indicated as follows:

Prerequisite

- ☑ This is a prerequisite that must be met before a complex action.

Action

1. This is a prompt.
2. This is a prompt. **CAUTION! This is a warning that notifies the user of hazards that may occur in combination with a given prompt.**
3. This is a prompt.
 - ⇒ This is the intermediate result of a given prompt.
 - ✓ This is the result of a complex action.

Result

Undefined action sequence

Complex actions that need not be carried out one after the other in a clear sequence are indicated as follows:

Prerequisite

- ☑ This is a prerequisite that must be met before a complex action.

Action

- ▶ This is a prompt.
- ▶ This is a prompt. **CAUTION! This is a warning that notifies the user of hazards that may occur in combination with a given prompt.**
- ▶ This is a prompt.
 - ⇒ This is the intermediate result of a given prompt.
 - ✓ This is the result of a complex action.

Result

2 Safety notes

2.1 Target user group

ZEISS AI IOL Calculator shall be used by trained health-care professionals such as: optometrists, ophthalmic technicians and ophthalmic surgeons.

The clinical decisions are made by the health-care professionals (not by ZEISS IOL Calculator) and are ultimately the responsibility of the professional based on his or her knowledge and experience. These decisions should follow accepted medical and surgical guidelines within the user's local practice environment and should always be based on a desire to meet the patient's needs and desires.

Trained ZEISS experts are involved for the acceptance and release of the trained, validated and tested parameter sets.

Rx Only

Caution: Federal (USA) Law restricts this device to sale by or on order of a physician.

2.2 Target patient group

The calculated results by ZEISS AI IOL Calculator are used by a surgery planning software for patients requiring intraocular lens (IOL) implantation due to cataract.

2.3 Area of use

2.3.1 Intended use

ZEISS AI IOL Calculator is an algorithm intended for the prediction of the post-operative residual refraction for a given eye biometry and IOL power.

2.3.2 Indications

ZEISS AI IOL Calculator is indicated for use in cataract surgery.

The output of ZEISS AI IOL Calculator is made available to be used by a surgery planning software.

2.3.3 Contraindications

No contraindications are known within the intended use.

2.4 Operator responsibilities and obligations

2.4.1 Operating staff

- Make sure you are thoroughly familiar with the contents of the instructions for use before putting the software into service. Also observe the instructions for use of any systems integrating the software.
- Keep the instructions for use readily available for users at all times.
- Familiarize yourself with the user settings of the software.
- Use the software for the intended purpose and as described only.
- The software must only be used by trained health-care personnel. Liability claims will be rendered void if the product is used by untrained personnel, and damage is incurred. Furthermore, this may result in a loss of possible warranty and guarantee claims.
- The product must not be modified. If you make changes to the product that are not described in these instructions for use, you are considered the responsible manufacturer of the device according to national legislations. Furthermore, this may result in a loss of possible warranty and guarantee claims.
- Please observe the statutory stipulations on accident prevention as well as occupational health and safety applicable in the corresponding country.
- Selecting the formula to be used as well as determining the target refraction and the corresponding IOL refractive power are the responsibility of the clinical staff and the attending physicians. Basic rules of medicine and the individual patient profile must be observed when doing so.

Safe functioning

The software is a support system and does not assume any diagnostic functions that are critical to safety. Treatments can be performed without the aid of the software at any time. Contact ZEISS Service in the event of a fault (see Troubleshooting ► 14)].

2.4.2 Messages to manufacturer and authorities

If a serious incident occurs in connection with this software affecting the operator or another person, the operator (or person responsible) must report this serious incident to the manufacturer and distributor of the software.

2.4.3 Data protection and information security

Information on data protection and information security of the software can be found in the instructions for use for the surgery planning software.

2.4.4 Place of installation and network

- The software may only be operated in secured networks. Appropriate security mechanisms must be installed on the computer systems (e.g., anti-virus programs), and the hardware may only be operated within an operator's environment that is compliant with IT guidelines. The operator is responsible for the safety and integrity of the software and network infrastructure. The consequences of attacks on the network cannot be predicted.
- The operator must take appropriate measures to ensure that only authorized users can access the computers on which the software is used (e.g., by locking the computer).
- Analysis, evaluation, and if necessary, risk management measures by the user/operator are recommended. These must be updated any time changes are made to the network or data connections.
- Data as well as settings generated by the software are included in the ZEISS AI IOL Calculator data backup. The operator of the software is responsible for backing up the data on a regular basis.

2.5 Patient information

Ensure that all residual risks, including all foreseeable, adverse events or side effects in connection with the use of the software or accessories, are explained to the patient. Note the following: Contraindications, safety information, warnings, limitations.

2.6 Product support period

The manufacturer guarantees support for this product for 3 years from the date of purchase.

3 Software description

3.1 Explanation of symbols

The following symbols are used on product labels or in these instructions for use:

	Warning
	Reference to electronic instructions for use (eIFU): www.zeiss.com/ifu Observe instructions for use
	Date of manufacture and country of manufacture
	Manufacturer
	UDI (unique device identification)
	Reference number
	Federal (USA) Law restricts this device to sale by or on order of a physician.

3.2 Conformity

The product does not qualify as a medical device according to current FDA legislations. However the product was developed under a certified quality management system according to ISO 13485.

Further legal information for the product such as manufacturer address, date of manufacture, and information pertaining to regulatory markings can be viewed via the surgery planning software.

3.3 Functional description

ZEISS AI IOL Calculator is an intraocular lens (IOL) calculation algorithm for cataract surgery and other anterior eye segment surgeries that uses AI methods^[1] (i.e., supervised machine learning) to predict the remaining spherical equivalent (SE) for a specific ocular biometry and IOL and IOL power.

The software is opened and controlled using surgery planning software. It is not necessary to install ZEISS AI IOL Calculator. Depending on the license of compatible surgery planning software, ZEISS AI IOL Calculator is automatically activated in the surgery planning software. Information on licensing the surgery planning software can be found in the surgery planning software instructions for use. Compatibility with Internet browsers or similar systems depends on the surgery planning software and can be found in the surgery planning software instructions for use. ZEISS AI IOL Calculator does not have its own user interface.

The patient's biometric data is entered or imported into the surgery planning software. The calculated IOLs are output via the surgery planning software.

Calculating the parameter sets

For each IOL that can be selected in the surgery planning software, a parameter set is initially calculated. The parameter sets are made available via the surgery planning software after being manually released. The parameter sets are optimized by ZEISS at random intervals (depending on the growth of the data) and then manually released again. How the parameter sets are calculated is described below.

1. The so-called "trainer" is filled with post- and preoperative data from real eyes. The provided data goes through quality assurance measures before being used for training purposes.
2. Using an AI model, a parameter set for a specific IOL is calculated from the data.
3. The parameter set is automatically tested using specific performance metrics. If the parameter set passes the test, it is checked and released by ZEISS. The user always has the current released parameter set automatically.

^[1] AI: Artificial intelligence

Calculating an IOL

How an IOL is calculated using ZEISS AI IOL Calculator is described below.

1. A patient's biometric data is entered or imported into the surgery planning software for IOL calculation.
2. Based on the biometric data, the selected IOL, the parameter set for the selected IOL, and the desired target refraction, the calculator calculates the IOL with the lowest residual refraction as well as nearby IOL powers with similar residual refraction.
3. The calculated IOLs are output to the surgery planning software.

Value ranges

ZEISS AI IOL Calculator is designed for inputs within the following value ranges. For inputs outside these value ranges, an error message is displayed in the surgery planning software. No calculation is made.

Input	Unit	Minimum value	Maximum value
Patient's sex	–	Male, female, unknown	
Patient's age	years	0	120
Axial length (AL)	mm	18.00	34.00
Anterior chamber depth (ACD)	mm	2.00	4.50
Lens thickness (LT)	mm	3.00	5.70
Central corneal thickness (CCT)	mm	0.40	0.70
Mean radius of curvature (flat)	mm	6.90	10.90
Mean radius of curvature (steep)	mm	6.50	10.90
Target refraction	D	-3.00	1.00
IOL power	D	-10.00	35.00

More information

For more information about ZEISS AI IOL Calculator, contact ZEISS Service.

4 Troubleshooting

Error messages are made available via the interface and displayed via the surgery planning software. You can find more information on how error messages are displayed in the surgery planning software instructions for use.

Error messages are displayed either as error code or as plain text. For all errors that you as a user cannot resolve yourself, please contact ZEISS Service. Please specify the error code or error message for this.

You can find the ZEISS representative for your country online on the following website: www.zeiss.com/med

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